Worksheet 5-5 & 5-2 Absolute Extrema #1
AP Calculus Name_____

Directions: Find the absolute minimum and absolute maximum values for the following functions on the given intervals.

1. $y = 3 - x \quad [-1, 2]$

2. $y = 5 - 2x^2 \quad [-3, 1]$

3. $g(x) = 2x^2 - 8x \quad [0, 6]$

4. $y = x^3 + \frac{3}{2}x^2 \quad [-1, 2]$

5. $y = \sqrt[3]{x} \quad [-8, 8]$

6. $f(t) = \dfrac{t}{t+3} \quad [-1, 6]$

7. $y = 2\sin(\theta) \quad [0, 2\pi]$

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9. A particle moves along the y-axis so that its velocity at time t, $0 \le t \le 6$, is given by $v(t) = 2(t-2)(t-5)$. Find the minimum velocity of the particle.

10. Find the maximum acceleration attained on the interval $0 \le t \le 3$ by the particle whose velocity is given by $v(t) = \frac{2}{3}t^3 - 4t^2 + 8t - 2$.